

VINOTH BALAJI

Technical Lead | Associate Manager | Full-Stack, Data & GenAI Engineering

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EXECUTIVE PROFILE

Technical Lead and Associate Manager with 8+ years of experience building enterprise software, analytics, and AI-powered products. Leads multiple concurrent client workstreams while remaining hands-on across Angular, TypeScript, Python, FastAPI, Node.js, Databricks, SQL, Azure, and GenAI. Trusted by stakeholders to turn ambiguous business problems into scalable solutions, manage delivery expectations, and constructively challenge requirements. Mentors engineers into independent feature owners and confident client-facing contributors.

CORE EXPERTISE

Leadership & Delivery: Technical Leadership, Solution Architecture, Client Engagement, Stakeholder Management, Team Mentoring, End-to-End Delivery

Engineering: Angular, TypeScript, Python, FastAPI, Node.js, Django, REST APIs, Highcharts, Redis, WebSockets

AI, Data & Analytics: Generative AI, Agentic AI, LLMs, LangGraph, Databricks, SQL, PostgreSQL, VaR/Risk Analytics, Power BI

Cloud & Security: Microsoft Azure, Entra ID, SSO/RBAC, JWT/OAuth, Docker, Kubernetes, Nginx, Application Security

CURRENT EXPERIENCE

ACCENTURE | Associate Manager / Technical Lead | Nov 2023 - Present

Houston, TX

- Lead end-to-end engineering and delivery across five concurrent enterprise products spanning Aviation, Pricing Analytics, Data Products, VaR/Risk Analytics, operational workflows, and Generative AI.
- Own requirements, solution design, frontend/backend engineering, data integration, security, testing, performance, production stabilization, and stakeholder acceptance rather than limiting ownership to a single layer.
- Partner directly with client stakeholders to clarify ambiguous asks, surface risks and dependencies, manage expectations, and recommend better technical or business approaches when appropriate.
- Mentor 4-5 engineers through design/code reviews, 1:1 coaching, and progressive ownership; developed team members into independent feature owners who can lead client discussions and make sound technical decisions.
- Drive AI-assisted engineering practices across development, analysis, troubleshooting, requirement refinement, and repetitive engineering work to improve delivery speed and team effectiveness.

Flagship Programs

- **Aviation Commercial Platform & AI Copilot:** Lead a greenfield commercial platform covering RFP, Bid, Contract, pricing, volume, and commercial decision workflows. Driving AI Copilot use cases for airport opportunity discovery, bid strategy, volume forecasting, margin analysis, tolerance risk, and negotiation intelligence.
- **Pricing Analytics Dashboard:** Lead a standalone enterprise pricing product with semantic instrument search, interactive dashboards, bi-directional filters, personalization, SSO, administration, subscriptions, responsive UX, and Databricks-backed analytics.
- **VaR Risk Analytics & Scenario Platform:** Lead exposure, position, curve, leg, and scenario analytics with multidimensional risk analysis, scenario modeling, complex visualization, data-quality controls, UOM/date validation, and high-performance analytical workflows.

CURRENT PORTFOLIO - ADDITIONAL PRODUCTS

- **CDP Excel Plugin & Web Application:** Drive a web + Excel enterprise data product enabling hierarchical filtering, Databricks-backed data access, downloads, persistent filters/metadata, refreshable workbooks, environment migration, and access-driven functionality.
- **MyOps - Work Order Visualizer, CATCH & RACE:** Lead operational enhancements across maintenance scheduling, optimization, action/case management, SAP-linked workflows, equipment data, saved/cascading filters, comments, and interactive schedule/network visualizations.

CROSS-PROGRAM ENGINEERING IMPACT

- Architect and deliver reusable patterns across authentication, authorization, API integration, user preferences, filtering, visualization, and enterprise data access instead of solving the same problem independently in each product.
- Work across Databricks, PostgreSQL, REST services, analytics layers, and frontend applications to trace issues end to end and resolve root causes across system boundaries.
- Apply performance engineering to data-heavy experiences through query optimization, efficient API/data-loading patterns, responsive visualization, and progressive UX/loading behavior.
- Bring GenAI into product strategy as well as engineering productivity - from Aviation commercial decision intelligence to AI-assisted development, troubleshooting, and requirement analysis.

LEADERSHIP & DELIVERY IMPACT

- Delivered committed milestones on time across multiple overlapping workstreams with zero critical escalations, while balancing new development, defects, stakeholder feedback, and production stabilization.
- Maintained strong first-time acceptance with minimal rework and low defect leakage by embedding validation, edge-case review, data-quality checks, and regression thinking throughout delivery.
- Groomed engineers into emerging leaders who can engage clients without hesitation, ask clarifying questions, explain tasks and trade-offs clearly, and respectfully challenge incorrect assumptions with evidence and alternatives.
- Created a high-ownership team culture through open communication, knowledge sharing, constructive feedback, and progressive delegation; reduced dependency on direct oversight while improving team throughput.
- Promoted responsible adoption of GenAI and AI-assisted development practices to accelerate engineering, analysis, troubleshooting, and requirement refinement while retaining strong review and quality controls.

EARLIER EXPERIENCE

FLUTURA DECISION SCIENCES & ANALYTICS | Oct 2017 - Nov 2023

Associate Technical Lead - Full Stack | Senior Programmer | Programmer

- Reduced Canvas/Workbook initial load time by 70% using lazy loading, on-demand loading, and container image optimization for data-intensive analytical workflows.
- Designed Node.js/Express middleware and secure Azure AD SSO using JWT/OAuth; conducted security audits and remediated vulnerabilities across enterprise applications.
- Led development of a config-driven Output Engine, reusable visualization/widget capabilities, and aggregation frameworks for interactive analysis of large datasets.
- Implemented real-time application capabilities with WebSockets and Redis; containerized services with Docker and built custom logging/usage analytics for reliability and easier debugging.
- Led legacy jQuery-to-MEAN modernization and mentored engineers on architecture, API optimization, debugging, code quality, and customer-facing feature development.

EDUCATION & RECOGNITION

Bachelor of Engineering (Computer Science) - Anna University, 2013 - 2017

Recognition: The Extra Mile Award

Selected Certifications: Agile & TDD Best Practices for APIs | OWASP Overview | Release & Sprint Planning | Kanban | Design Thinking